

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON ACADEMIC MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by

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JAVA ASSIGNMENT

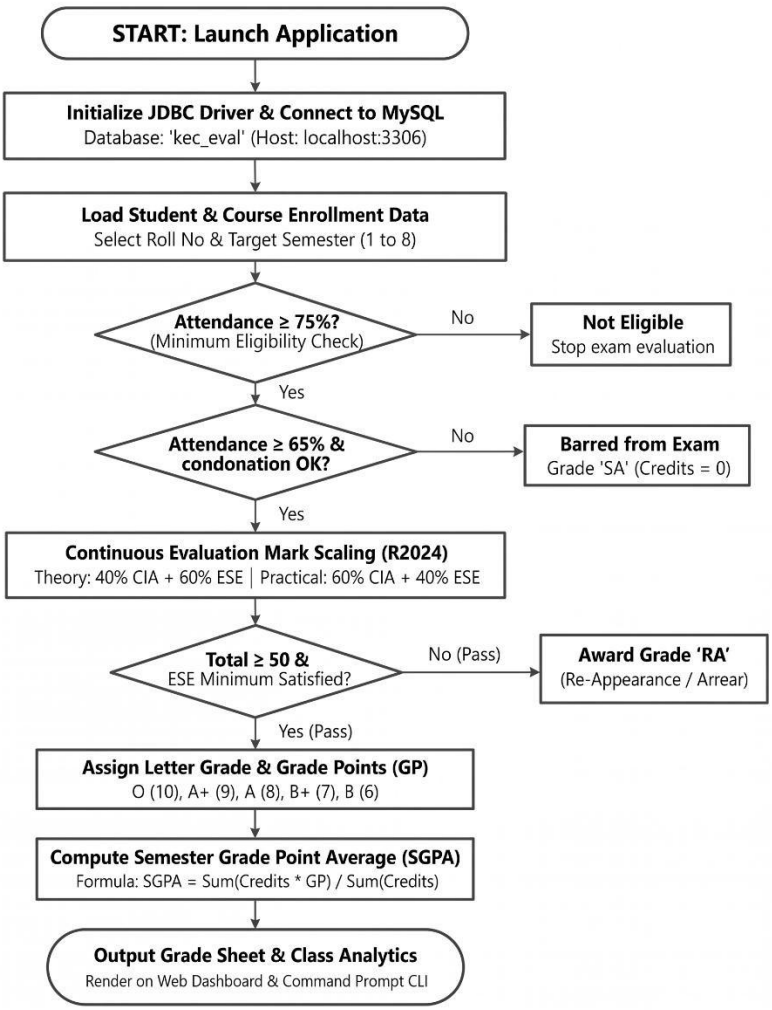
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

Field	Type	Null	Key	Default	Extra
roll_number	TEXT	NO	PRI	NULL	
name	TEXT	NO		NULL	
department	TEXT	NO	MUL	NULL	
current_semester	INTEGER	NO		NULL	

2. COURSES TABLE:

Field	Type	Null	Key	Default	Extra
course_code	TEXT	NO	PRI	NULL	
title	TEXT	NO		NULL	
credits	REAL	NO		NULL	
course_type	TEXT	NO		NULL	

3. ENROLLMENTS TABLE:

Field	Type	Null	Key	Default	Extra
enrollment_id	INTEGER	NO	PRI	NULL	auto_increment
roll_number	TEXT	NO	MUL	NULL	
course_code	TEXT	NO	MUL	NULL	
semester	INTEGER	NO	MUL	NULL	
total_classes	INTEGER	NO		0	
attended_classes	INTEGER	NO		0	
medical_condonation_approved	INTEGER	NO		0	

4. ASSESSMENT MARKS TABLE:

Field	Type	Null	Key	Default	Extra
enrollment_id	INTEGER	NO	PRI	NULL	
cia_test1	REAL	YES		NULL	
cia_test2	REAL	YES		NULL	
assignment	REAL	YES		NULL	
ese_mark	REAL	YES		NULL	
ese_absent	INTEGER	NO		0	

SOURCE CODE:

MAINAPP.JAVA:

```
package edu.kec.academic;


import java.sql.SQLException;

import java.util.*;


public class MainApp {

   Scanner sc = new

Scanner(System.in);

   StudentDAO studentDAO =

new StudentDAO();

   AcademicDAO academicDAO

= new AcademicDAO();
```

```
    public static void
main(String[] args) {
        try {

DBConnection.initializeDatabas
e();
            new MainApp().run();
        } catch (SQLException e) {

System.out.println("Error: " +
e.getMessage());
        }
    }

    void run() {
        while (true) {

System.out.println("\nKEC
R2024 Academic System");

System.out.println("1.Student
2.Course  3.Enroll");
```

```
System.out.println("4.Attendanc
e  5.CIA  6.Condonation");
        System.out.println("7.ESE
8.Mark Sheet  9.Defaulters");

System.out.println("10.Departm
ent Summary  0.Exit");

        String ch =
input("Choice");
        try {
            switch (ch) {
                case "1" ->
studentDAO.addStudent(
                    input("Roll"),
input("Name"),

input("Department"),
number("Semester"));
                case "2" ->
academicDAO.addCourse(
                    input("Course
Code"), input("Title"),
```

```
                    decimal("Credits"),

AcademicDAO.CourseType.value
Of(

input("Type").toUpperCase()));
                case "3" ->
academicDAO.enrollStudent(
                    input("Roll"),
input("Course Code"),

number("Semester"));
                case "4" ->
academicDAO.recordAttendanc
e(
                    input("Roll"),
input("Course Code"),

number("Semester"),
number("Classes Held"),
                    number("Classes
Attended"));
                case "5" ->
academicDAO.recordCiaMark(
```

```
                    input("Roll"),
input("Course Code"),

number("Semester"),
number("Component"),
                    decimal("Mark"));
                case "6" ->
academicDAO.approveMedicalC
ondonation(
                    input("Roll"),
input("Course Code"),

number("Semester"));
                case "7" -> {
                    String roll =
input("Roll");
                    String course =
input("Course Code");
                    int sem =
number("Semester");
                    boolean absent =
input("Absent?
y/n").equalsIgnoreCase("y");
```

```
                    Double mark =
absent ? null : decimal("ESE
Mark");

academicDAO.recordEseMark(r
oll, course, sem, mark, absent);
                }
                case "8" ->
markSheet();
                case "9" ->
defaulters();
                case "10" ->
summary();
                case "0" -> {

System.out.println("Goodbye!")
;
                    return;
                }
                default ->
System.out.println("Invalid
choice");
            }
```

```
            } catch (SQLException |
IllegalArgumentException e) {

System.out.println("Error: " +
e.getMessage());
            }
        }
    }

    void markSheet() throws
SQLException {
        String roll = input("Roll");
        int sem =
number("Semester");
        var rows =
academicDAO.getMarkSheet(rol
l, sem);

        for (var r : rows)
            System.out.println(r);

        if (!rows.isEmpty()) {
```

```
            double sgpa =
AcademicDAO.calculateSgpa(ro
ws);
            System.out.printf("SGPA:
%.2f%nPercentage: %.2f%%%n",
                    sgpa, sgpa * 10);
        }
    }

    void defaulters() throws
SQLException {
        var rows =
academicDAO.getAttendanceDe
faulters(number("Semester"));

rows.forEach(System.out::printl
n);
        if (rows.isEmpty())
System.out.println("No
defaulters found.");
    }

    void summary() throws
SQLException {
```

```
        var rows =
academicDAO.getDepartmentSu
mmary(number("Semester"));

rows.forEach(System.out::printl
n);
        if (rows.isEmpty())
System.out.println("No results
found.");
    }

    String input(String msg) {
        System.out.print(msg + ":
");
        return sc.nextLine().trim();
    }

    int number(String msg) {
        return
Integer.parseInt(input(msg));}
    double decimal(String msg) {
        return
Double.parseDouble(input(msg)
);
```

DB CONNECTION.JAVA:

```
package edu.kec.academic;


import java.io.BufferedReader;

import java.io.IOException;

import java.io.InputStream;

import java.io.InputStreamReader;

import java.nio.charset.StandardCharsets;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;

import java.sql.Statement;


public final class DBConnection {

   private static final String URL =

System.getProperty("kec.db.url", "jdbc:sqlite:kec_academic.db");


   private DBConnection() {

   }


   public static Connection getConnection() throws SQLException

{

       try {

           Class.forName("org.sqlite.JDBC");

       } catch (ClassNotFoundException exception) {

           throw new SQLException("SQLite JDBC driver is

missing. Run the application with Maven.", exception);
```

```
        }
        Connection connection =
DriverManager.getConnection(URL);
        try (Statement statement = connection.createStatement()) {
            statement.execute("PRAGMA foreign_keys = ON");
        }
        return connection;
    }

    public static void initializeDatabase() throws SQLException {
        try (InputStream input =
DBConnection.class.getResourceAsStream("/kec_schema.sql")) {
            if (input == null) {
                throw new SQLException("kec_schema.sql was not
found on the application classpath.");
            }
            StringBuilder sql = new StringBuilder();
            try (BufferedReader reader = new BufferedReader(new
InputStreamReader(input, StandardCharsets.UTF_8));
                 Connection connection = getConnection();
                 Statement statement = connection.createStatement()) {
                String line;
                while ((line = reader.readLine()) != null) {
                    sql.append(line).append('\n');
                }
                for (String command : sql.toString().split(";")) {
                    if (!command.isBlank()) {
                        statement.execute(command);
```

```
                }

            }

        } catch (IOException exception) {

            throw new SQLException("Could not read
kec_schema.sql.", exception);

        }

    } catch (IOException exception) {

        throw new SQLException("Could not close
kec_schema.sql.", exception);

}
```

```
ublic static String getPassword() {
        return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") :
DEFAULT_PASS;
    }

    static {
try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found in classpath: " + e.getMessage());
        }
    }

    /**
* Obtains a new database connection
     */
    public static Connection getConnection() throws SQLException {
return DriverManager.getConnection(getUrl(), getUser(), getPassword());
    }

    /**
* Tests the database connectivity
     */
    public static boolean testConnection() {
try (Connection conn = getConnection()) {
return conn != null && !conn.isClosed();
        }
```

OUTPUT SCREENSHOTS:

MAIN MENU:

STUDENT & COURSE MANAGEMENT:

```
KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

1.STUDENT REGISTRATION:

```
Select an option: 1
Roll number: 25ITR120
Name: SAHANA
Department: INFORMATION TECHNOLOGY
Current semester: 3
Student registered.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

2. COURSE CATALOG:

```
Select an option: 2
Course code: IT033
Course title: JAVA PROGRAMMING
Credits: 3
Course type (THEORY, PRACTICAL, EMBEDDED): THEORY
Course added.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
Select an option: 3
Roll number: 25ITR120
Course code: IT032
Semester: 3
Student enrolled.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

```
Select an option: 4
Roll number: 25ITR120
Course code: IT032
Semester: 3
Classes held in this update: 10
Classes attended in this update: 10
Attendance updated.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
Select an option: 5
Roll number: 25ITR120
Course code: IT032
Semester: 3
Component (1 = CIA Test 1, 2 = CIA Test 2, 3 = Assignment / Quiz / Lab): 1
Mark (0-100): 80
CIA mark saved.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
Select an option: 5
Roll number: 25ITR120
Course code: IT032
Semester: 3
Component (1 = CIA Test 1, 2 = CIA Test 2, 3 = Assignment / Quiz / Lab): 2
Mark (0-100): 85
CIA mark saved.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

3. PHASE 3 (POST SEMESTER):

```
Select an option: 5
Roll number: 25ITR120
Course code: IT032
Semester: 3
Component (1 = CIA Test 1, 2 = CIA Test 2, 3 = Assignment / Quiz / Lab): 3
Mark (0-100): 90
CIA mark saved.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
Select an option: 7
Roll number: 25ITR120
Course code: IT032
Semester: 3
Absent in ESE? (y/n): n
ESE mark (0-100): 80
ESE result saved.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

STUDENT SEMESTER MARK SHEET:

```
Select an option: 8
Roll number: 25ITR120
Semester: 3

Mark sheet: 25ITR120, Semester 3
Course       Title                  CIA1 / CIA2 / Asg   Internal  ESE      External  Total    Grade Status

IT032        FSD                    80.00 / 85.00 / 90.00 34.00     80.00    48.00     82.00    A+    PASS

SGPA: 9.00 | Equivalent percentage: 90.00%

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
Select an option: 9
Semester: 3

Attendance defaulters (below 65%, grade SA)
Roll         Name                   Department   Course       Attendance
No attendance defaulters found.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

```
Select an option: 10
Semester: 3

Department result summary (students with unresolved results are excluded)
Department       Passed   Failed   Average SGPA
INFORMATION TECHNOLOGY 2        0        9.00

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

```
Select an option: 6
Roll number: 25ITR120
Course code: IT032
Semester: 3
Medical condonation approval recorded.

KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option:
```

```
Select an option: 0
Goodbye.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  39:21 min
[INFO] Finished at: 2026-10-24T20:28:32+05:30
[INFO] ------------------------------------------------------------------------
```

DATABASE:

STUDENTS DATABASE:

enrollment_id	roll_number	course_code	semester	total_classes	attended_classes	medical_condonation_approved
1	25ITR163	IT031	3	10	10	0
2	25ITR165	IT032	3	0	0	0
3	25ITR120	IT032	3	10	10	1

COURSES DATABASE:

enrollment_id	cia_test1	cia_test2	assignment	ese_mark	ese_absent
1	89.0	85.0	90.0	80.0	0
3	80.0	85.0	90.0	80.0	0

ENROLLMENTS DATABASE:

```
KEC R2024 Student Academic & Continuous Evaluation System
1. Register student  2. Add course  3. Enroll student
4. Record attendance  5. Enter CIA mark  6. Approve medical condonation
7. Enter ESE mark / absence  8. Student semester mark sheet
9. Attendance defaulters  10. Department result summary  0. Exit
Select an option: 10
Semester: 3

Department result summary (students with unresolved results are excluded)
Department       Passed   Failed   Average SGPA
INFORMATION TECHNOLOGY 2        0        9.00
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK:	https://github.com/MurasoliB/Student_Evaluation_Using_Java_JDBC